



Department of Telecommunications
Telecommunication Engineering Centre
K.L. Bhawan, Janpath, New Delhi – 110001
(Convergence & Broadcasting Division)



No: - 4-16/2018-C&B/TEC/NWG-9

dated: 25.11.2022

Subject: Minutes of the Sixth meeting of the National Working Group 9 (NWG-9), corresponding to ITU-T Study Group 9 (SG9) for the Study Period 2022- 2024, held on 21st November 2022.

The sixth meeting of the National Working Group-9 (NWG-9) corresponding to the ITU- T Study Group 9 (SG9) reconstituted for the Study Period 2022-24 regarding "Audio visual content transmission and integrated broadband cable networks" or "Broadband Cable and TV" was held on 21-11-2022 under the chairmanship of Sh. Avinash Agarwal, DDG (Convergence & Broadcasting), TEC through video-conferencing. List of participants is placed at Annexure-I.

2. At the outset, Sh. Rakesh Desai, Director (C&B-II) TEC & Convener of NWG-9, welcomed all the participants and briefly explained about the discussions held during the previous meetings. He briefly informed the members about the Agenda of the meeting and the work done by the working teams on the ongoing work items of NWG-9 in respect of contributions furnished to ITU-T SG-9.

3. Shri Avinash Agarwal, Chairman NWG-9 & DDG (C&B) TEC, in his opening remarks, stated that work of NWG-9 Members, who are engaged at various management positions in ITU-T SG9, such as Vice-chairman, Rapporteur, Associate Rapporteur, Editor, etc, is aplicated at ITU-T. He stated that the contributions from NWG-9 members in ITU-T need to be enhanced. He emphasised that members should study all the questions of SG9, identify the work items that are relevant for India, and work upon submitting more contributions in ITU-T SG9 to address the India specific concerns and to ensure wider acceptance of Indian standards globally.

4. Shri Avinash Agarwal expressed concerns that many members are not attending the NWG9 meetings regularly. He also requested all the members to identify relevant experts and nominate subject experts/ academia/practitioners as members of the group.

5. Sh. Rakesh Desai, Director, TEC briefly explained all eleven questions of ITU-T SG9 and their terms of references to the group members. He also briefly explained the latest WTSA (World Telecommunication Standardization Assembly) resolutions and requested the members to examine the resolutions and propose amendments or new resolutions pertaining to SG9. He pointed out that India has proposed to host the next WTSA to be held in 2024.

6. Sh. Rakesh Desai also highlighted further developments in the Standards Coordination Portal of TEC and demonstrated new features like chatbot, outreach, email subscription, etc. to the members. He also demonstrated how to register as portal member, upload contributions, and get access of the information related to all activities of NWGs on the portal. He stated that this portal is at present under testing and will be ready by the end of this month. He said that once the portal is ready, members of NWGs will be requested to enroll on the portal tentatively w.e.f 1st December 2022 and be familiar with various functions and features of the portal. All the members appreciated the concept of Standards Coordination Portal and found the functions of the portal very useful in sharing the information and collaborating on the standards development activities.

7. Prof. Dr. Pradipta Biswas, IISc Bengaluru & Vice-Chair SG9/NWG-9, stated that he is working on contribution regarding “*Common user profile format for audio-visual content distribution*” and on accessibility aspect with Sh. Avinash Agarwal. He stated that the members have to finalise their contributions in r/o ongoing recommendations / technical reports latest by February 2023 in view of forthcoming meeting of SG9 in March/ April 2023. He assured that he will nominate members from academia who are interested to involve actively in NWG9 activities.
8. Shri Manavdeep Singh, C-DOT explained the progress in work-item in respect of Q1/9- ‘*J.TR.WiFiTV (Technical Report) on Secondary distribution of digital television and audiovisual content to portable devices using Wi-Fi*’. He stated that the working team is still updating the draft and will finalise the draft expeditiously. Sh. Avinash Agarwal informed that this contribution will be discussed in the forthcoming Q1/9 Rapporteur meeting dated 13/12/2022 and we have to submit our updated contribution by 6th December 2022 and therefore the working team may please ensure to finalise their draft by 30/11/2022.
9. Sh. Rajiv Khattar explained the progress in work-item TR-FSR (Technical report) regarding ‘*Factual subscriber base reporting and protected content delivery in conditional access system*’. He stated that he is further updating the draft and sought suggestions / inputs from the members on this work item.
10. Ms. Meenakshi Singhvi briefly explained Q6/9 of ITU-T SG9- ‘*Functional requirements for terminal devices of the integrated broadband cable network*’ and presented the progress in work item ‘J.BD2M’ regarding “*Functional requirements for terminal devices of the Broadcast Direct to Mobile (D2M) networks*”. She stated that this work item is being discussed in the forthcoming Q6/9 Rapporteur meeting wherein she and the Sh. Avinash Agarwal will explore how this contribution remains within the scope of SG9 terms of reference.
11. Sh. Rakesh Desai briefly explained Q8/9 of ITU-T SG9- ‘*The Internet protocol (IP) enabled multimedia applications and services for cable television networks enabled by converged platforms*’ and discussed the two ongoing contributions submitted by other member-state regarding draft recommendations on - Requirements/ Specifications of microservice architecture for audio-visual media in the converged media cloud. He stated that ITU-T SG16 is also working on similar topic pertaining to IPTV application infrastructure based on microservices. He requested all the members concerned from academia and industry to furnish their inputs / comments on these work items.
12. Sh. Avinash Agarwal briefly explained Q11/9 of ITU-T SG9- ‘*Accessibility to cable systems and services*’ and apprised the group on the progress in ongoing work-item regarding ‘*JSTR.LCAP (Technical Report) Technical advances, challenges, and best practices in live captioning*’. He complimented the sincere efforts of the working team of this work item in providing inputs for the draft of the technical report. Sh. Shubhanshu Gupta, on behalf of CDAC-Pune team, explained the drafts of Chapter 6- ‘Captioning Standards’ and Chapter 8- ‘Recent Advances’. Sh.Desai presented draft prepared by BIF team (Ms.Neema and Dr.Nirmita) on Chapetrer-7- ‘Quality And Accuracy Measurement’. Dr. Ashay Sahasrabudhe explained the drafts prepared by him on Chapter 9- ‘Use Cases and Chapter 10- ‘Case studies and global best practices’. Sh. Avinash Agarwal informed that Q11/9 Rapporteur meeting is scheduled on 01/12/2022 and he further asked the participants for their suggestions / feedback on these drafts. Dr. Sahasrabudhe was requested to provide inputs on case studies and global best practices at the earliest.
13. On request of the Chair regarding nominations of experts as members of this group, Major General Dr. Pritam Bishnoi, TSDSI responded that more than 100 participants attended the recently concluded TSDSI Tech Deep Dive Conference 2022 and that he would share the list of experts, who

are working in the relevant areas of NWG9, shortly with the group.

14. Sh. Sundar Narayanan referred the Q3/9 - *AI-enabled enhanced functions over integrated broadband cable network* and suggested to work on the adoption of AI.

15. Sh. Avinash Agarwal, in his concluding remarks, highlighted the significance of the Standards Coordination portal of TEC stating that it will be a good repository of work done by the members and will serve as a knowledge resource of high-quality standards and technical reports in the areas of information and communication technologies (ICTs) for members and non-members as well as for students and researchers. He stressed upon to expand the group by adding more active experts and therefore, he again requested the members to suggest the names of experts from OEMs, Industry, Association, Start-ups, academia, practitioners, research scholars, etc.

16. The key action points emerged from the meeting are as under: -

- a) The Working Teams may submit their drafts by **30th November 2022**.
- b) Members may submit their comments on draft technical report on live captioning by **30th November 2022**.
- c) Members may suggest the names of experts from OEMs/ Industry/ Association/ Start-ups/ academia/ practitioners/ students/ research scholars that can be added in the group.
- d) Members may examine the WTSA (World Telecommunication Standardization Assembly) resolutions and propose amendments or new resolutions pertaining to SG9.
- e) Members may study all the questions of ITU-T SG9 and come forward to increase number of Indian contributions in ITU-T SG9.

17. The meeting ended with the vote of thanks to the chair.

R. S. Desai

(Rakesh Desai)

Convener, NWG-9 & Director, (C&B-II), TEC

Encl.: Annexure-I

To,
All members of the NWG-9 and all the Participants.

Copy for kind information to:

1. Sr.DDG & Head TEC, New Delhi

ANNEXURE-I

List of Participants:

1. Sh. Avinash Agarwal, Chairman NWG-9 & DDG (C&B), TEC.
2. Dr. Pradipta Biswas, IISc Bengaluru & Vice-Chair NWG-9.
3. Ms. Meenakshi Singhvi, Prasar Bharti.
4. Sh. Rajiv Khattar, Researcher & Industry specialist
5. Major General Dr. Pritam Bisnoi, TSDSI.
6. Sh. Subhanshu Gupta, C-DAC, Pune
7. Sh. Rajith Ali- SIA India
8. Dr. Ashay Sahasrabudhe
9. Sh. Manav C-DoT
10. Sh. Sundar Narayanan, Researcher & Consultant
11. Sh. Shiv
12. Sh. Sumeet Ahuja
13. Sh. Sunil Gautam, Director (SPPI), DoT.
14. Sh. Rakesh Desai, Convener, NWG-9 & Director (C&B)-II, TEC.
15. Sh. Brajesh Kumar, Director (C&B)-I, TEC.
16. Ms. Muskan Chaudhary, Intern, TEC.
17. Mr Adil Shaharyar, Intern, TEC.
